

Karthik D K

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Summary

Firmware Developer building RISC-V software ecosystems for automotive and IoT at Bosch's TrusteD-V innovation team. 6+ years of embedded systems experience spanning kernel development, RTOS evaluation, and automotive ECU firmware. Expertise in Rust and C for bare-metal and safety-critical systems on ARM Cortex-M and RISC-V architectures.

Experience

Senior Software Engineer	Bosch Global Software Technologies	August 2019 - Present
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TrusteD-V: RISC-V Rust Software Ecosystem

- Core developer on Bosch's TrusteD-V innovation team, building a secure RISC-V software ecosystem targeting automotive and IoT applications.
- Ported Hubris OS to RISC-V architecture, implementing kernel primitives, trap/exception handlers, and PMP-based memory isolation for Hazard3 RISC-V core on Raspberry Pi Pico 2 board.
- Overcame hardware-specific challenges including RP2350 errata workarounds and Physical Memory Protection configuration for secure multi-tasking.
- Debugged kernel bring-up issues using Segger J-Link with OpenOCD/GDB on Mindgrove Secure IoT development board.
- Work featured at RISC-V Summit 2025: "Unlocking the Potential of RISC-V With TrusteD-V"

MLlib DecisionTree Inference

- Developed an inference algorithm for DecisionTree in C for PowerTrain ECU.
- Utilized Flat Buffers for defining and parameterizing the model through calibrations.

Advanced Exhaust Temperature Management

- Developed a comprehensive component for reading temperature sensors via CAN and implementing sensor diagnostics to comply with EU7 standards.
- Implemented statistical analysis to monitor temperature distribution across catalyst, improving emission control efficiency.

Predictive Emission Modelling for Motorcycles

- Conceptualized and implemented machine learning models to predict the emissions of nitrogen oxides (NOx), hydrocarbons (HC), and carbon monoxide (CO) in motorcycles, addressing critical challenges in vehicle emissions.
- Achieved superior R² scores and secured the runner-up position in a Bosch India AI hackathon, demonstrating predictive accuracy and expertise in emission data analytics.

Skills

Programming: C, Rust, Python

RTOS/Bare-Metal: Hubris OS, FreeRTOS, Zephyr, bare-metal programming

Debug Tools: Segger J-Link, RP Debug Probe, OpenOCD, GDB, JTAG/SWD debugging

Developer Tools and Frameworks: Cargo, Make, Git, Just, GCC, ASCET, PyTorch, Linux

Certifications

Machine Learning Coursera	Certificate 2020
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Achievements

- RISC-V Summit 2025:** TrusteD-V project contributor (featured talk). 
- Bosch Recognition:** Best Developer Award, Crowdsourcing Champ, AI Hackathon Winner. 
- Chess:** State-Level Championship participant.

Education

University Vishweshwaraya College of Engineering	75.2%
B.E in Electronics and Communication	2019
Devaraja URS PU College	94.5%
PUC	2015

Blogging

- Documenting my knowledge with programming, software engineering and motorcycles to create a knowledge repository for future reference, aiding both emerging LLMs and my personal growth. 

Open-Source Contributions

- Actively contributing to GitHub projects, notable contributions to fastai and FluxML Deep Learning libraries. 

Projects

Open Source Projects

exhubris

- Configured linker scripts and startup code for dual-architecture Hubris OS demo on RP Pico 2W.
- Handled architecture-specific boot sequences and vector tables for both ARM [Cortex-M33](#) and RISC-V [Hazard3](#) cores.
- Validated hardware bring-up using GDB over SWD with RP Debug Probe.

robot-hat-rs

- Developed no_std Rust drivers for I2C, PWM, and ADC peripherals targeting the Robot HAT robotics platform.
- Implemented hardware abstraction layer for motor control, servo actuation, and analog sensor reading.
- Published on [crates.io](#), receiving thousands of downloads from the community.

ixv

- Developed a CLI application in Rust for verifying Intel HEX file(s).
- Published the binary on [crates.io](#), receiving thousands of downloads from the community.

picars

- Created an autonomous vehicle system using Raspberry Pi and PiCar-X kit, leveraging Rust and Python.
- Developed Rust bindings and interfaced with Python to optimize execution speed.

rprs

- Developed a CLI application in Rust for replacing file(s).

GeekyMicky

- Tackled the Micro-Mouse Maze Challenge using Arduino Uno and C programming.

Graduation Project

Closed-Loop Control of Anesthesia Administration

- Designed an automated closed-loop control system for General Anesthesia using a PID controller.
- Developed a system to regulate the depth of hypnosis using propofol administration and Bi-Spectral Index (BIS) as a controlled variable.
- Implemented the project using MATLAB/Simulink.

Languages

- English
- Kannada
- Telugu